Here is a step-by-step C5.0 decision tree example using the **Weather/Golf dataset**, with detailed calculations and explanations of C5.0's enhancements over C4.5:

Step 1: Dataset Overview

14 instances, 4 features, binary target:

Day	**Outlook**	**Temperature**	**Humidity**	**Windy**	**Play Golf?**
D1	Sunny	Hot	High	False	No
D2	Sunny	Hot	High	True	No
D3	Overcast	Hot	High	False	Yes
D4	Rainy	Mild	High	False	Yes
D5	Rainy	Cool	Normal	False	Yes
D6	Rainy	Cool	Normal	True	No
D7	Overcast	Cool	Normal	True	Yes
D8	Sunny	Mild	High	False	No
D9	Sunny	Cool	Normal	False	Yes
D10	Rainy	Mild	Normal	False	Yes
D11	Sunny	Mild	Normal	True	Yes
D12	Overcast	Mild	High	True	Yes

D13	Overcast	Hot	Normal	False	Yes
D14	Rainy	Mild	High	True	No

Target distribution: Yes = 9, No = 5

Key C5.0 features: Automatic continuous feature handling, boosting support, rule generation

Step 2: Root Node Selection (Gain Ratio)

2.1 Total Entropy $E(D)$:

$$E(D) = -\left(\frac{9}{14}\log_2\frac{9}{14} + \frac{5}{14}\log_2\frac{5}{14}\right) = 0.940$$

Measures initial uncertainty in "Play Golf?"

2.2 Information Gain (IG) Calculation:

Outlook (categorical):

Sunny (5): 2 Yes, 3 No → $E = 0.971$

Overcast (4): 4 Yes → $E = 0$

Rainy (5): 3 Yes, 2 No → $E = 0.971$

$$IG = 0.940 - \left(\frac{5}{14} \times 0.971 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.971\right) = 0.246$$

Humidity (continuous, auto-discretized):

Best split: **Humidity ≤ 75**

≤75 (6): All Yes → $E = 0$

> 75 (8): 3 Yes, 5 No → $E = 0.954$

$$IG = 0.940 - \left(\frac{6}{14} \times 0 + \frac{8}{14} \times 0.954\right) = 0.151$$

Other features:

Windy: IG = 0.048

Temperature: IG = 0.029

2.3 Gain Ratio (C5.0's Key Metric):

$$GainRatio(A) = \frac{IG(A)}{SplitInfo(A)}$$

Outlook:

$$SplitInfo = -\left(\frac{5}{14}\log_2\frac{5}{14} + \frac{4}{14}\log_2\frac{4}{14} + \frac{5}{14}\log_2\frac{5}{14}\right) = 1.577$$

$$GainRatio = 0.246/1.577 = 0.156$$

Humidity:

$$GainRatio = 0.151/0.985 = 0.153$$

Root node = Outlook (highest gain ratio)

Step 3: Build Sub-Trees

3.1 Branch: Outlook = Sunny (5 instances)

Subset entropy: $E = 0.971$

Optimal split: Humidity (GainRatio = 0.153)

Humidity ≤ 75: **Play=Yes** (2 instances)

Humidity > 75: **Play=No** (3 instances)

3.2 Branch: Outlook = Overcast (4 instances)

All "Yes" → **Leaf node (Play=Yes)**

3.3 Branch: Outlook = Rainy (5 instances)

Subset entropy: $E = 0.971$

Optimal split: Windy (GainRatio = 1.0)

Windy = False: **Play=Yes** (3 instances)

Windy = True: **Play=No** (2 instances)

Step 4: Final Decision Tree

```
Outlook
 ├── Sunny → Humidity
|     ├── ≤75 → Play=Yes
|    └── >75 → Play=No
 ├── Overcast → Play=Yes
```

```
└── Rainy → Windy
    ├── False → Play=Yes
    └── True → Play=No
```

Temperature excluded (lowest gain ratio)

C5.0 Enhancements Over C4.5

1.**Automatic Continuous Feature Handling**:

●Finds optimal splits for continuous features (e.g., **Humidity ≤ 75**) without manual intervention.

2.**Boosting Support**:

●Builds multiple trees with weighted voting (e.g., 10-tree ensemble) to improve accuracy.

3.**Rule Generation**:

Converts trees to simplified rules:

Rule 1: IF Outlook=Overcast THEN Play=Yes
Rule 2: IF Outlook=Sunny AND Humidity≤75 THEN Play=Yes
Rule 3: IF Outlook=Rainy AND Windy=False THEN Play=Yes
ELSE Play=No

4.**Efficiency Optimizations**:

●100x faster than C4.5 on large datasets.

●90% less memory usage.

Prediction Example

Input: [Outlook=Rainy, Humidity=80, Windy=False]

Path:

1.Outlook → Rainy branch

2.Windy → False branch

Output: **Play=Yes** (Matches instances D4, D5, D10)

Key Formulas

Concept	Formula
Entropy	$E(S) = -\sum(p_i \log_2 p_i)$
Information Gain	$IG(A) = E(S) - \sum \frac{\lvert S_v \rvert}{\lvert S \rvert} E(S_v)$
Split Information	$SplitInfo(A) = -\sum \frac{\lvert S_v \rvert}{\lvert S \rvert} \log_2 \frac{\lvert S_v \rvert}{\lvert S \rvert}$
Gain Ratio	$GainRatio(A) = \frac{IG(A)}{SplitInfo(A)}$

C5.0 maintains interpretability while significantly boosting efficiency and accuracy through these innovations.